

WEEK1 SUMMARY

Range of the data = Max. Value – Min value

FREQUENCY DISTRIBUTION

Time (min)	Frequency (f)	Time (min)	Frequency (f)
2	1	14	4
4	1	15	6
5	2	16	3
6	1	17	3
7	1	19	2
8	2	21	1
9	2	23	1
10	3	24	1
11	4	25	1
12	3	26	1
13	5	28	1
		31	1

Another way of showing the frequency of values in a data set is using a **frequency distribution (FD) table**.

$$\sum_i f_i = n = 50$$

RELATIVE FREQUENCY DISTRIBUTION

The **relative frequency** of a class $i, i=1,2,...,k$, can be computed as:

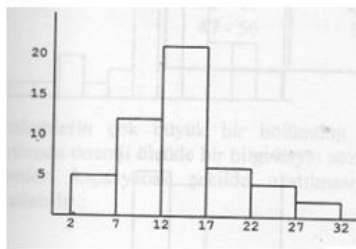
$$\text{Relative Frequency } (f_i^R) = \frac{\text{Frequency } (f_i)}{\text{Number of data } (n)}$$

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)
2-7	4,5	5	5/50=0,10
7-12	9,5	12	12/50=0,24
12-17	14,5	21	21/50=0,42
17-22	19,5	6	6/50=0,12
22-27	24,5	4	4/50=0,08
27-32	29,5	2	2/50=0,04

$$\sum_i f_i^R = 1$$

HISTOGRAMS

frequency



Time (min)

Steps for Creating CFD:

1. Compute the range of data.
2. Determine the number of classes (k). In practice, k can be chosen generally between 5-15. You may use the following formulas to specify it:

$$k = \sqrt{n}$$

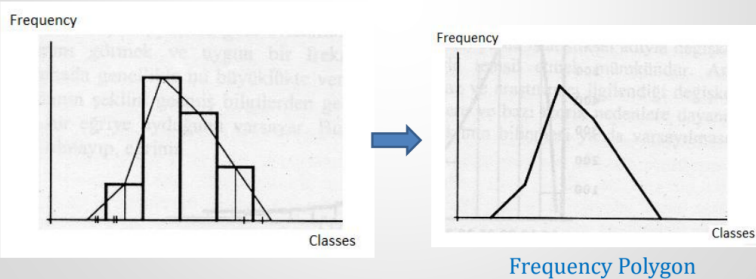
$$k = 1 + 3.3 \log n$$

$$k = 5 \log n$$

3. Divide the data into classes with equal intervals. To create these classes, first compute the class interval:

$$\text{Class interval} = \left[\frac{\text{Range of data}}{\text{Number of classes } (k)} \right]$$

4. Find the midpoints (x_i) of each class.
5. Compute the frequency (f_i) of each class.

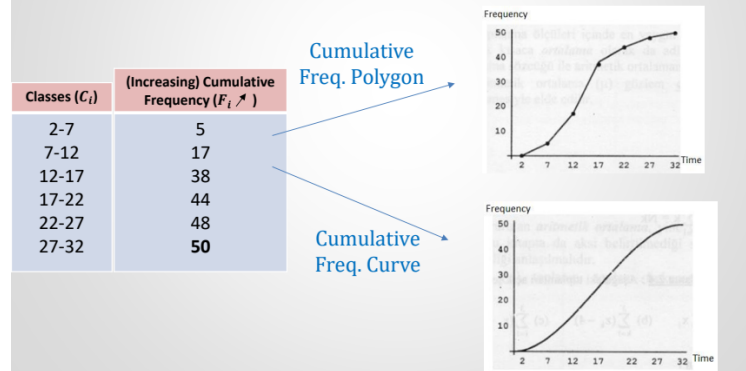


Frequency Polygon

(INCREASING) CUMULATIVE FREQUENCY DISTRIBUTIONS

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)	(Increasing) Cumulative Frequency ($F_i^{\nearrow}$)	Cumulative Relative Frequency ($F_i^{R\nearrow}$)
2-7	4,5	5	5/50=0,10	5	0,10
7-12	9,5	12	12/50=0,24	17	0,34
12-17	14,5	21	21/50=0,42	38	0,76
17-22	19,5	6	6/50=0,12	44	0,88
22-27	24,5	4	4/50=0,08	48	0,96
27-32	29,5	2	2/50=0,04	50	1,00

CUMULATIVE FREQUENCY POLYGON AND CURVE



(DECREASING) CUMULATIVE FREQUENCY DISTRIBUTION

Classes (C_i)	Midpoints (x_i)	Frequency (f_i)	Relative Frequency (f_i^R)	(Decreasing) Cumulative Frequency ($F_i^{\searrow}$)	Cumulative Relative Frequency ($F_i^{R\searrow}$)
2-7	4,5	5	5/50=0,10	50	1,00
7-12	9,5	12	12/50=0,24	45	0,90
12-17	14,5	21	21/50=0,42	33	0,66
17-22	19,5	6	6/50=0,12	12	0,24
22-27	24,5	4	4/50=0,08	6	0,12
27-32	29,5	2	2/50=0,04	2	0,04

FREQUENCY CURVE

